

VARAD PATIL

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI/ML and Full-Stack Engineer with a first-author Springer publication on RAG systems and 2+ years building production web applications. Skilled in LLM/RAG pipelines, deep learning for forecasting, and MERN/Next.js development on AWS; improved performance and automated workflows by 30–40% for enterprise clients.

EDUCATION

Indian Institute of Technology Kanpur — M.Tech, AI for Sustainability 2025–2027
GPA: 9.25/10 | Machine Learning, Deep Learning, Sustainable AI Systems Kanpur, India

Rajarambapu Institute of Technology — B.Tech, Computer Engineering 2021–2025
GPA: 9.70/10 | Data Structures, Algorithms, DBMS, Web Technologies, Software Engineering Maharashtra, India

RESEARCH PUBLICATION

A RAG-Based Legal Chatbot Using Llama2 and Advanced Information Retrieval 2024
First Author — Proceedings of ICTCS 2024, Springer Python, LangChain, FAISS, Llama2, Flask, React

- Engineered an end-to-end RAG pipeline with FAISS vector indexing and Llama2-7B via LangChain (corpus chunking, embeddings, prompt templates).
- Built an evaluation framework measuring precision, latency, and user satisfaction across 500+ test queries.

PROFESSIONAL EXPERIENCE

Software Development Intern — Altair Engineering Jan 2025 – Jul 2025
Bangalore, India React, Node.js, Express, MongoDB, Python, AWS

- Built full-stack MERN client portals for CEAT, MRF, and Mahindra with HPC job scheduling, real-time monitoring, and visualization dashboards.
- Automated simulation pipelines using Python and REST APIs, reducing manual operations by 40%.
- Improved frontend performance by 30% through code-splitting, lazy loading, and API caching strategies.
- Implemented AWS integrations (S3, EC2, Athena) for secure, scalable storage and compute.

Web Developer — Bubble Byte Ventures Pvt Ltd Sep 2022 – Aug 2023
Sangli, India React, JavaScript, Python, REST APIs

- Delivered 10+ responsive, SEO-optimized client websites, cutting page load times by 30% via asset optimization and caching.
- Automated deployment operations with Python scripts; shipped cross-browser, mobile-responsive interfaces.

PROJECTS

Active-Break Indian Summer Monsoon Forecasting | PyTorch, TCN, Transformers, XGBoost

- Built a deep learning framework classifying Active, Break, and Normal monsoon phases up to 7 days ahead from ERA5 reanalysis data.
- Developed a TCN–Transformer meta-ensemble that outperformed persistence and climatological baselines.

Cyclone Track Prediction System | XGBoost, Scikit-learn, Python

- Engineered 40+ IBTrACS features (motion vectors, intensity gradients, spatial context) to predict cyclone positions at 6h–72h.
- Achieved 14.47 km RMSE at 6h and 78.77 km at 24h with geodesic metrics and skill-score analysis vs. a persistence baseline.

Full-Stack Web Applications | React, Node.js, Express, Flask, MongoDB, Stripe, Gemini API

- AI Resume Analyzer:** Gemini-powered semantic analysis with ATS scoring, keyword matching, and MongoDB-backed history.
- E-Commerce Platform:** JWT auth, role-based access control, order tracking, admin dashboard, and secure Stripe payments.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

AI/ML: RAG Systems, LangChain, FAISS, Llama2, HuggingFace, Prompt Engineering, Vector Embeddings, NLP

Deep Learning: PyTorch, TensorFlow, Transformers, TCN, scikit-learn, XGBoost, Time-Series Forecasting

Frontend: React.js, Next.js, Redux, Tailwind CSS, Framer Motion, Responsive UI

Backend & Databases: Node.js, Express.js, FastAPI, Flask, REST APIs, MongoDB, PostgreSQL, MySQL, Vector DBs

Cloud & Tools: AWS (S3, EC2, Lambda, Athena), GCP, Docker, Git, GitHub, Vercel, Postman

ACHIEVEMENTS & LEADERSHIP

Reliance Foundation Postgraduate Scholarship (2025): awarded to 100 postgraduate scholars nationwide |

GATE 2025: All India Rank 722.

Community Lead, Google Developer Groups (GDG) RIT (2024–2025): led 45+ members, ran 5+ workshops on React and GenAI, mentored 20+ students.

Certifications: Generative AI with LLMs (DeepLearning.AI); Programming Foundations (Duke University).